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TYPE OF SLIDING DOOR RAILING

Abstract

The present invention discloses a new type of sliding door railing and, during installation, the first connecting mechanism and the second connecting mechanism are spaced in opposite direction fat the installation position, and then the opposite sides of door railing body are detachably and pivotally connected to the first connecting mechanism and second connecting mechanism, respectively, the connection between the door railing body and the first connecting mechanism can be disconnected, and the door railing body can be forced to swing for door opening and closing, with the connection position between the door railing body and the second connecting mechanism as the rotation center; the connection between the door railing body and the second connecting mechanism can also be disconnected.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Patent Application No. 202420440248.9, filed on Mar. 7, 2024, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present invention relates to the field of door railing technology, in particular to a new type of sliding door railing.

BACKGROUND

[0003] In families with infants, in order to protect infants walking out of the room against accidents such as getting lost or personal injury, people often install a sliding door railing in positions like indoor aisles, and the sliding door railing is generally composed of two movably connected door bodies, one of the door bodies can be slid to lengthen or shorten the door railing, and during installation, the length of aisle can be first adjusted according to the length of aisle, and one side of the door railing can be pivotally installed in the installation position, so that the door railing can swing between the position where the aisle is disconnected and the position where the aisle is open and, when the door railing swings to the position where the aisle is disconnected, the other side of the door railing can be locked by the door railing lock. Since the pivot structure on one side is used to swing the door railing and the locking structure on the other side is used to fix the door railing, this type of sliding door railing is a single-side opening door, and the sliding door railings with monotonous structure and operation can hardly arouse consumer attention and interest.

SUMMARY

[0004] The main purpose of the present invention is to propose a new type of sliding door railing to address the problems proposed in the background art mentioned above.

[0005] To achieve the above purpose, the present invention proposes a new type of sliding door railing, including a door railing body, a first connecting mechanism, and a second connecting mechanism, the first connecting mechanism and the second connecting mechanism are fixed at intervals in opposite in the installation position, and one side of the door railing body is detachably and pivotally connected to the first connecting mechanism, and the side of the door railing body far away from the first connecting mechanism is detachably and pivotally connected to the second connecting mechanism.

[0006] The present invention discloses a new type of sliding door railing and, during installation, the first connecting mechanism and the second connecting mechanism are spaced in opposite direction at the installation position, and then the opposite sides of door railing body are detachably and pivotally connected to the first connecting mechanism and second connecting mechanism, respectively, the connection between the door railing body and the first connecting mechanism can be disconnected, and the door railing body can be forced to swing for door opening and closing, with the connection position between the door railing body and the second connecting mechanism as the rotation center; the connection between the door railing body and the second connecting mechanism can also be disconnected, and the door railing body can be forced to swing for door opening and closing, with the connection position between the door railing body and the first connecting mechanism as the rotation center. Compared with traditional methods, the new type of sliding door railing having a relatively simple structure can be quickly removed when needed, providing ideal practicality and novelty.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. 1 is a 3D view of the door railing body, first connecting mechanism, and second connecting mechanism during assembly;

[0008] FIG. 2 is a 3D view of the door railing body, first connecting mechanism, and third connecting mechanism;

[0009] FIG. 3 is a cross-sectional view of the first connecting mechanism;

[0010] FIG. 4 is a cross-sectional view of the second connecting mechanism.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0011] The technical solutions in the embodiments of the present invention will be described clearly and completely in conjunction with the accompanying drawings of the embodiments of the present invention. Obviously, the described embodiments are only a part of the embodiments of the present invention, rather than all the embodiments. Based on the embodiments of the present invention, all other embodiments obtained by those of ordinary skill in the art without creative work shall fall within the protection scope of the present invention.

[0012] It should be noted that all directional indicators (such as up, down, left, right, front, back, top, bottom, inside, outside, vertical, horizontal, vertical, counterclockwise, clockwise, circumferential, radial, axial . . .) in the embodiments of the present invention are only used to explain the relative position relationship and movement among various components in a particular posture (as shown in the accompanying drawings), and if that particular posture is changed, the directional indications will change accordingly.

[0013] In addition, descriptions involving terms “first,” “second” and the like, if any, in the present invention are only for illustrative purposes and cannot be understood as indicating or implying relative importance or the number of features referred to. Therefore, the features defined with “first” and “second” may explicitly or implicitly include at least one or a plurality of the features. At the same time, the technical solutions of various embodiments can be combined with each other, but must be based on what those of ordinary skill in the art can achieve. The combination of technical solutions that result in contradiction or make it impossible to implement should be considered non-existent, and accordingly should not fall within the scope of protection required by this present invention.

[0014] The present invention proposes a new type of sliding door railing.

[0015] In the embodiments of the invention, as shown in FIGS. 1 to 4, a new type of sliding door railing, including a door railing body 1, a first connecting mechanism 2, and a second connecting mechanism 3, the first connecting mechanism 2 and the second connecting mechanism 3 are fixed at intervals in opposite in the installation position, one side of the door railing body 1 is detachably and pivotally connected to the first connecting mechanism 2, and the side of the door railing body 1 far away from the first connecting mechanism 2 is detachably and pivotally connected to the second connecting mechanism 3.

[0016] The present invention discloses a new type of sliding door railing and, during installation, the first connecting mechanism and the second connecting mechanism are spaced in opposite direction at the installation position, and then the opposite sides of door railing body are detachably and pivotally connected to the first connecting mechanism and second connecting mechanism, respectively, the connection between the door railing body and the first connecting mechanism can be disconnected, and the door railing body can be forced to swing for door opening and closing, with the connection position between the door railing body and the second connecting mechanism as the rotation center; the connection between the door railing body and the second connecting mechanism can also be disconnected, and the door railing body can be forced to swing for door opening and closing, with the connection position between the door railing body and the first connecting mechanism as the rotation center. Compared with traditional methods, the new type of sliding door railing having a relatively simple structure can be quickly removed when needed,

providing ideal practicality and novelty.

[0017] Specifically, the upper left end and lower left end of the door railing body are detachably and pivotally connected to the upper end and lower end of the first connecting mechanism, and the upper right end and lower right end of the door railing body are detachably and pivotally connected to the upper end and lower end of the second connecting mechanism, respectively.

[0018] In further, the first connecting mechanism **2** includes a first upper seat **21** and a first lower seat **22** that have a first push button **23**, the bottom of the first upper seat **21** and the top of the first lower seat **22** are respectively provided with a first socket **211** and a second socket **221**, and a first expansion member **11** is movably provided at the position where the door railing body **1** is assembled with the first upper seat **21**, the first expansion member **11** can, relative to the door railing body **1**, move between the position where it partially extends into the first socket **211** and the position where it exits the first socket **211**, and a first top spring (not shown in figure) is additionally provided to apply elastic force to cause the first expansion member **11** move in the direction in which the first expansion member **11** can partially extend into the first socket **211**;

[0019] a second expansion member **12** is movably installed at the position where the first lower seat **22** is assembled with the door railing body **1**, the second expansion member **12** can, relative to the door railing body **1**, move between the position where it partially extends into the second socket **221** and the position where it exits the second socket **211**, and a second top spring is additionally provided to apply elastic force to cause the second expansion member **12** move in the direction in which the second expansion member **12** can partially extend into the second socket **221**, and the door railing body can maintain detachable pivot fit with the first connecting mechanism through the cooperation between the first expansion member and the first socket, as well as the cooperation between the second expansion member and the second socket; [0020] the first socket **211** extends to the top of the first upper seat **21** that runs through the first upper seat **21**, and a first pressure piece **24** is movably installed on the upper end of the first socket **211**, the first pressure piece **24** can perform vertical reciprocating motion along the first socket **211** to force the first expansion piece **11** to exit the first socket **211** or move away from the first expansion piece **11**, the first push button **23** can move, relative to the first upper seat **21**, between the position where it locks the first pressure piece **24** and the position where it unlocks the first pressure piece **24**, and the top of door railing body can be disconnected from the first upper seat by pressing down on the first pressure rod and, during operation, the first push button should be first driven to unlock the first pressure rod to avoid the disassembly of the door railing body due to accidental touch; [0021] the second connecting mechanism **3** includes a second upper seat **31** and a second lower seat **32** that have a second push button **33**, the bottom of the second upper seat **31** and the top of the second lower seat **32** are respectively provided with a third socket **311** and a fourth socket **321**, a third expansion member **13** is movably installed at the position where the second upper seat **31** is assembled with the door railing body **1**, the third expansion member **13** can, relative to the door railing body **1**, move between the position where it partially extends into the third socket **311** and the position where it exits the third socket **311**, and a sixth top spring is additionally provided to apply elastic force to cause the third expansion member **13** move in the direction in which the third expansion member **12** can partially extend into the third socket **311**; [0022] a fourth expansion member **14** is movably installed at the position where the second lower seat **32** is assembled with the door railing body **1**, the fourth expansion member **14** can, relative to the door railing body **1**, move between the position where it partially extends into the fourth socket **321** and the position where it exits the fourth socket **321**, and a seventh top spring is additionally provided to apply elastic force to cause the fourth expansion member **14** move in the direction in which the fourth expansion member **14** can partially extend into the fourth socket **321**, the door railing body can maintain detachable pivot fit with the second connecting mechanism through the combination of the third expansion member and the third socket, as well as the combination of the fourth expansion member and the fourth socket; [0023] the third socket **311** extends to the top of the second upper seat **31**, and a second

pressure piece **34** is movably installed on the upper end of the third socket **311**, the second pressure piece **34** can perform vertical reciprocating motion along the third socket **311** to force the third expansion piece **13** to exit the third socket **311** or move away from the third expansion piece **13**, the second push button **33** can, relative to the second upper seat **31**, move between the position where it locks the second pressure piece **34** and the position where it unlocks the second pressure piece **34**, and the top of door railing body can be disconnected from the second upper seat by pressing down on the second pressure rod and, during operation, the second push button should be first driven to unlock the first pressure rod to avoid the disassembly of the door railing body due to accidental touch.

[0024] Specifically, a first upper seat **21** is internally provided with a first follow-up portion **25** that can move with the first pressure piece **24**, the first upper seat is internally and movably provided with a first slider **26** that is connected to the first push button **23**, and a first strip groove **261** that runs through the top and bottom walls of the first slider **26** is provided on the side near the first pressure piece **24**, the first slider **26** can move between the position where the first follow-up portion **25** aligns with the upper opening of the first strip groove **261** and the position where the first follow-up portion **25** is away from the upper opening of the first strip groove **261**, a first top piece **27** is movably provided inside the first upper seat **21** at a position below the first slider **26**, and when the first follow-up portion **25** aligns with the opening at the upper end of the first strip groove **261**, the first top piece **27** can move between the position where it partially extends into the first strip groove **261** and the position where it exits the first strip groove **261**, and a third top spring **28** is additionally provided to apply elastic force to cause the first slider **26** move in the direction in which the first follow-up portion **25** is away from the upper opening of the first strip groove **261**, and a fourth top spring **29** is additionally provided to apply elastic force to cause the first top piece **27** move in the direction in which the first top piece **27** can partially extend into the first strip groove **261**; the first top piece **27** maintains elastic contact with the bottom wall of the first slider **26** through the elastic force applied by the fourth top spring **29** and, when the first slider **26** moves to the position where the first follow-up portion **25** aligns with the upper opening of the first strip groove **261**, the elastic force applied by the fourth top spring **29** can force the upper end of the first top piece **27** to extend into the first strip groove **261** to fix the first slider **26**. When the first pressure piece **24** moves to a position where the first expansion piece **11** is forced to exit the first socket **211**, the first follower portion **25** moves to force the first top piece **27** to exit the first strip groove **261** to unlock the first slider **26** and, after the first slider **26** is unlocked, the third spring forces the first slider **26** to move in the direction close to the first pressure piece **24**, so that the first top piece is staggered from the first strip groove **261** and maintains elastic contact with the bottom wall of the first slider **26**, and the first follow-up portion **25** maintains elastic contact with the first strip groove **261** and, when the first pressure piece **24** goes up (for example, lifting the first pressure piece **24** by external force or during installation, pushing the first pressure piece **24** upward by the first expansion piece **11**), the first follow-up portion **25** is driven to exit the first strip groove **261**, so that the first slider **26** can move towards the direction where the upper opening of the first strip groove **261** is staggered from the first follow-up portion **25**, and the first pressure piece **24** can be locked.

[0025] Specifically, a first push member **222** is movably provided inside the first lower seat **22**, first push member **222** can move between the position where it partially extends into the second socket **221** and the position where it exits the second socket **221**, and a fifth top spring is additionally provided to apply elastic force to cause the first push member **222** move in the direction in which the first push member **222** partially, an inclined plane, extends into the second socket **221**, and when the second expansion member **12** extends into the second socket **221**, the inclined plane can force the portion that has been inserted into the second socket **221** to exit.

[0026] Specifically, the top of the first push member **222** is provided with a first mark (not shown in the figure) and a second mark (not shown in the figure), and the top of the first lower seat **22** is

provided with a first window **223** and, after the first push member **222** partially extends into the second socket **221**, the first mark is aligned up and down with the first window **223** and, after the first push member **222** partially exits the second socket **221**, the second mark is aligned up and down with the first window **223**. In this embodiment, the first mark indicates that the second expansion member **12** is not inserted into the second socket **221**, and the second mark indicates that the second expansion member **12** has been inserted into the second socket **221**, ensuring that the door railing body **1** is assembled with the first lower seat **22**.

[0027] Specifically, a first support rod **41** is provided, and the two ends of the first support rod **41** are respectively connected with the first upper seat **21** and the first lower seat **22**, the first support rod **41** is movably provided with a first moving seat **42**, and the side of the first moving seat **42** away from the first door body is provided with a first plug board that can be inserted into or withdrawn from the first moving seat **42**, the first plug board is fixed in the installation position, and the first support rod **41** can be fixed in the installation position through the connection between the first moving seat **42** and the first plug board, suitable for the installation position with a skirting line.

[0028] Specifically, the first upper seat **21** is detachably fixed in the installation position through the first quick release mechanism, and reference can also be made to prior art for the first quick release mechanism, which will not be further elaborated here.

[0029] Specifically, the second upper seat **31** is internally provided with a second follow-up portion that can move with the second pressure piece **34**, a second slider **36** that is connected to the second push button **33** is movably provided inside the second upper seat **31**, and a second strip groove that runs through the top and bottom walls of the second slider **36** is provided on one side of the second slider **34** near the second pressure piece **34**, the second slider **36** can move between the position where the second follow-up portion is aligned with the upper opening of the second strip groove and the position where the second follow-up portion is away from the the upper opening of the second strip groove, and a second top piece **37** is movably provided inside the second upper seat **31** corresponding to the position below the second slider **36**, and when the second follow-up portion aligns with the upper opening of the second strip groove, the second top member **37** can move between the position where it partially extends into the second strip groove and the position where it exits the second strip groove, an eighth top spring **38** is additionally provided to apply elastic force to cause the second slider **36** move in the direction in which the second follow-up portion is away from the the upper opening of the second strip groove, and a ninth top spring **39** is also provided to apply elastic force to cause the second top piece **37** move in the direction in which it partially extends into the second strip groove. The second top piece **37** maintains elastic contact with the bottom wall of the second slider **36** through the elastic force applied by the ninth top spring **39** and, when the second slider **36** moves to the position where the second follow-up portion aligns with the upper opening of the second strip groove, the elastic force applied by the ninth top spring **39** can force the upper end of the second top piece **37** to extend into the second strip groove to fix the second slider **36**. When the second pressure piece **34** moves to a position where the second expansion piece **12** is forced to exit the third socket **311**, the second follower portion moves to force the second top piece **37** to exit the second strip groove to unlock the second slider **36** and, after the second slider **36** is unlocked, the eighth spring **38** forces the second slider **36** to move in the direction close to the second pressure piece **34**, so that the second top spring is staggered from the first strip groove and maintains elastic contact with the bottom wall of the second slider **36**, and the second follow-up portion maintains elastic contact with the second strip groove and, when the second pressure piece **34** goes up (for example, lifting the second pressure piece **34** by external force or during installation, pushing the second pressure piece **34** upward by the second expansion piece **12**), the second follow-up portion is driven to exit the second strip groove, so that the second slider **36** can move towards the direction where the upper opening of the second strip groove is staggered from the second follow-up portion, and the second pressure piece **34** can be locked.

[0030] Specifically, a second push member **322** is movably provided inside the second lower seat **32**, and the second push member **322** can move between the position where it partially extends into the fourth socket **321** and the position where it exits the fourth socket **321**, and additionally a tenth top spring is provided to apply elastic force to cause the second push member **322** move in the direction where it partially, an inclined plane, extends into the fourth socket **321**, and when the fourth expansion member **14** extends into the fourth socket **321**, the inclined plane can force the portion that has been inserted into the fourth socket **321** to exit.

[0031] Specifically, the top of the second push member **322** is provided with a third mark (not shown in the figure) and a fourth mark (not shown in the figure), and the top of the second lower seat **32** is provided with a second window and, after the second push member **322** partially extends into the fourth socket **321**, the third mark is aligned up and down with the second window and, after the second push member **322** partially exits the fourth socket **321**, the fourth mark is aligned up and down with the second window. In this embodiment, the third mark indicates that the fourth expansion member **14** is not inserted into the fourth socket **321**, and the fourth mark indicates that the fourth expansion member **14** has been inserted into the fourth socket **321**, ensuring that the door railing body **1** is assembled with the second lower seat **32**.

[0032] Specifically, a second support rod **51** is provided, and the two ends of the second support rod **51** are respectively connected with the second upper seat **31** and the second lower seat **32**, the second support rod **51** is movably provided with a second moving seat **52**, and the side of the second moving seat **52** away from the second door body is provided with a second plug board (not shown in the figure) that can be inserted into or withdrawn from the second moving seat **52**, the second plug board is fixed in the installation position, and the second support rod **51** can be fixed in the installation position through the connection between the second moving seat **52** and the second plug board, suitable for the installation position with a skirting line.

[0033] Specifically, the second upper seat **31** is detachably fixed in the installation position through the second quick release mechanism, and reference can also be made to prior art for the second quick release mechanism, which will not be further elaborated here.

[0034] Specifically, the door body is an integrated structure, or the door railing body **1** includes a first door body and a second door body, the top and bottom of the first door body, as well as the top and bottom of the second door body, are fixedly connected to a first bar member **15** and a second bar member **16**, respectively, the two ends of the first bar member **15** are bent and extended to cover partial back wall of the first door body and partial front wall of the second door body, and the two ends of the second bar member **16** are bent and extended to respectively cover partial back wall of the first door body and partial front wall of the second door body, and the second door body can, relative to the first door body, move between the position where the first strip members (**15**) are in contact with each other and the position where the first strip members (**15**) are away from each other, and the two bent and extending ends of the first strip member **15** and of the second strip member **16** can prevent the disconnection between the first and second door bodies; [0035] one side of the first door body is detachably and pivotally connected to the first connecting mechanism **2**, and the side of the second door railing away from the first door body is detachably and pivotally connected to the second connecting mechanism **3**, and the connection between the first door body and the first connecting mechanism **2** can be disconnected, and the door railing can swing and slide to restore the access, with the connection position between the second door body and the second connecting mechanism **3** as the rotation center; the connection between the second door body and the second connecting mechanism **3** can also be disconnected, and the door railing can swing and slide to restore the access, with the connection position between the first door body and the first connecting mechanism **2** as the rotation center.

[0036] Specifically, the first and second door bodies are integrated structures or foldable structures, and reference can also be made to prior art for the first quick release mechanism, which will not be further elaborated here.

[0037] Specifically, the bottom of the first door body and/or second door body is provided with one or more universal balls (not shown in the figure) for rolling contact with the ground.

[0038] The above are only the preferred embodiments of the present invention, and are not intended to limit the present invention. Any equivalent structural transformation made based on the description and accompanying drawings hereof under the concept of present invention, or be directly/indirectly applied in other related technical fields, should fall within scope of patent protection of the present invention.

Claims

1. A new type of sliding door railing, including a door railing body (1), a first connecting mechanism (2), and a second connecting mechanism (3), the first connecting mechanism (2) and the second connecting mechanism (3) are fixed at intervals in opposite in the installation position, one side of the door railing body (1) is detachably and pivotally connected to the first connecting mechanism (2), and the side of the door railing body (1) far away from the first connecting mechanism (2) is detachably and pivotally connected to the second connecting mechanism (3).
2. The new type of sliding door railing according to claim 1, wherein the upper left end and lower left end of the door railing body are detachably and pivotally connected to the upper end and lower end of the first connecting mechanism, and the upper right end and lower right end of the door railing body are detachably and pivotally connected to the upper end and lower end of the second connecting mechanism, respectively.
3. The new type of sliding door railing according to claim 2, wherein the first connecting mechanism (2) includes a first upper seat (21) and a first lower seat (22) that have a first push button (23), the bottom of the first upper seat (21) and the top of the first lower seat (22) are respectively provided with a first socket (211) and a second socket (221), and a first expansion member (11) is movably provided at the position where the door railing body (1) is assembled with the first upper seat (21), the first expansion member (11) can, relative to the door railing body (1), move between the position where it partially extends into the first socket (211) and the position where it exits the first socket (211), and a first top spring is additionally provided to apply elastic force to cause the first expansion member (11) move in the direction in which the first expansion member (11) can partially extend into the first socket (211); a second expansion member (12) is movably installed at the position where the first lower seat (22) is assembled with the door railing body (1), the second expansion member (12) can, relative to the door railing body (1), move between the position where it partially extends into the second socket (221) and the position where it exits the second socket (221), and a second top spring is additionally provided to apply elastic force to cause the second expansion member (12) move in the direction in which the second expansion member (12) can partially extend into the second socket (221); the first socket (211) extends to the top of the first upper seat (21) that runs through the first upper seat (21), and a first pressure piece (24) is movably installed on the upper end of the first socket (211), the first pressure piece (24) can perform vertical reciprocating motion along the first socket (211) to force the first expansion piece (11) to exit the first socket (211) or move away from the first expansion piece (11), the first push button (23) can move, relative to the first upper seat (21), between the position where it locks the first pressure piece (24) and the position where it unlocks the first pressure piece (24); the second connecting mechanism (3) includes a second upper seat (31) and a second lower seat (32) that have a second push button (33), the bottom of the second upper seat (31) and the top of the second lower seat (32) are respectively provided with a third socket (311) and a fourth socket (321), a third expansion member (13) is movably installed at the position where the second upper seat (31) is assembled with the door railing body (1), the third expansion member (13) can, relative to the door railing body (1), move between the position where it partially extends into the third socket (311) and the position where it exits the third socket (311), and a sixth top spring is

additionally provided to apply elastic force to cause the third expansion member (13) move in the direction in which the third expansion member (12) can partially extend into the third socket (311); a fourth expansion member (14) is movably installed at the position where the second lower seat (32) is assembled with the door railing body (1), the fourth expansion member (14) can, relative to the door railing body (1), move between the position where it partially extends into the fourth socket (321) and the position where it exits the fourth socket (321), and a seventh top spring is additionally provided to apply elastic force to cause the fourth expansion member (14) move in the direction in which the fourth expansion member (14) can partially extend into the fourth socket (321); the third socket (311) extends to the top of the second upper seat (31), and a second pressure piece (34) is movably installed on the upper end of the third socket (311), the second pressure piece (34) can perform vertical reciprocating motion along the third socket (311) to force the third expansion piece (13) to exit the third socket (311) or move away from the third expansion piece (13), the second push button (33) can, relative to the second upper seat (31), move between the position where it locks the second pressure piece (34) and the position where it unlocks the second pressure piece (34).

4. The new type of sliding door railing according to claim 3, wherein a first upper seat (21) is internally provided with a first follow-up portion (25) that can move with the first pressure piece (24), the first upper seat is internally and movably provided with a first slider (26) that is connected to the first push button (23), and a first strip groove (261) that runs through the top and bottom walls of the first slider (26) is provided on the side near the first pressure piece (24), the first slider (26) can move between the position where the first follow-up portion (25) aligns with the upper opening of the first strip groove (261) and the position where the first follow-up portion (25) is away from the upper opening of the first strip groove (261), a first top piece (27) is movably provided inside the first upper seat (21) at a position below the first slider (26), and when the first follow-up portion (25) aligns with the opening at the upper end of the first strip groove (261), the first top piece (27) can move between the position where it partially extends into the first strip groove (261) and the position where it exits the first strip groove (261), and a third top spring (28) is additionally provided to apply elastic force to cause the first slider (26) move in the direction in which the first follow-up portion (25) is away from the upper opening of the first strip groove (261), and a fourth top spring (29) is additionally provided to apply elastic force to cause the first top piece (27) move in the direction in which the first top piece (27) can partially extend into the first strip groove (261);

5. The new type of sliding door railing according to claim 3, wherein a first push member (222) is movably provided inside the first lower seat (22), first push member (222) can move between the position where it partially extends into the second socket (221) and the position where it exits the second socket (221), and a fifth top spring is additionally provided to apply elastic force to cause the first push member (222) move in the direction in which the first push member (222) partially, an inclined plane, extends into the second socket (221), and when the second expansion member (12) extends into the second socket (221), the inclined plane can force the portion that has been inserted into the second socket (221) to exit.

6. The new type of sliding door railing according to claim 5, wherein the top of the first push member (222) is provided with a first mark and a second mark, and the top of the first lower seat (22) is provided with a first window (223) and, after the first push member (222) partially extends into the second socket (221), the first mark is aligned up and down with the first window (223) and, after the first push member (222) partially exits the second socket (221), the second mark is aligned up and down with the first window (223).

7. The new sliding door railing according to claim 3, wherein a first support rod (41) is provided, and the two ends of the first support rod (41) are respectively connected with the first upper seat (21) and the first lower seat (22), the first support rod (41) is movably provided with a first moving seat (42), and the side of the first moving seat (42) away from the first door body is provided with a

first plug board that can be inserted into or withdrawn from the first moving seat (42), the first plug board is fixed in the installation position, and the first support rod (41) can be fixed in the installation position through the connection between the first moving seat (42) and the first plug board.

8. The new type of sliding door railing according to claim 3, wherein the second upper seat (31) is internally provided with a second follow-up portion that can move with the second pressure piece (34), a second slider (36) that is connected to the second push button (33) is movably provided inside the second upper seat (31), and a second strip groove that runs through the top and bottom walls of the second slider (36) is provided on one side of the second slider (36) near the second pressure piece (34), the second slider (36) can move between the position where the second follow-up portion is aligned with the upper opening of the second strip groove and the position where the second follow-up portion is away from the the upper opening of the second strip groove, and a second top piece (37) is movably provided inside the second upper seat (31) corresponding to the position below the second slider (36), and when the second follow-up portion aligns with the upper opening of the second strip groove, the second top member (37) can move between the position where it partially extends into the second strip groove and the position where it exits the second strip groove, an eighth top spring (38) is additionally provided to apply elastic force to cause the second slider (36) move in the direction in which the second follow-up portion is away from the the upper opening of the second strip groove, and a ninth top spring (39) is also provided to apply elastic force to cause the second top piece (37) move in the direction in which it partially extends into the second strip groove.

9. The new type of sliding door railing according to claim 3, wherein a second push member (322) is movably provided inside the second lower seat (32), and the second push member (322) can move between the position where it partially extends into the fourth socket (321) and the position where it exits the fourth socket (321), and additionally a tenth top spring is provided to apply elastic force to cause the second push member (322) move in the direction where it partially, an inclined plane, extends into the fourth socket (321), and when the fourth expansion member (14) extends into the fourth socket (321), the inclined plane can force the portion that has been inserted into the fourth socket (321) to exit.

10. The new type of sliding door railing according to claim 9, wherein the top of the second push member (322) is provided with a third mark and a fourth mark, and the top of the second lower seat (32) is provided with a second window and, after the second push member (322) partially extends into the fourth socket (321), the third mark is aligned up and down with the second window and, after the second push member (322) partially exits the fourth socket (321), the fourth mark is aligned up and down with the second window.

11. The new type of sliding door railing according to claim 3, a second support rod (51) is provided, and the two ends of the second support rod (51) are respectively connected with the second upper seat (31) and the second lower seat (32), the second support rod (51) is movably provided with a second moving seat (52), and the side of the second moving seat (52) away from the second door body is provided with a second plug board that can be inserted into or withdrawn from the second moving seat (52), the second plug board is fixed in the installation position, and the second support rod (51) can be fixed in the installation position through the connection between the second moving seat (52) and the second plug board.

12. The new type of sliding door railing according to claims 1, wherein the door railing body (1) includes a first door body and a second door body, the top and bottom of the first door body, as well as the top and bottom of the second door body, are fixedly connected to a first bar member (15) and a second bar member (16), respectively, the two ends of the first bar member (15) are bent and extended to cover partial back wall of the first door body and partial front wall of the second door body, and the two ends of the second bar member (16) are bent and extended to respectively cover partial back wall of the first door body and partial front wall of the second door body, and the

second door body can, relative to the first door body, move between the position where the first strip members (15) are in contact with each other and the position where the first strip members (15) are away from each other. one side of the first door body is detachably and pivotally connected to the first connecting mechanism (2), and the side of the second door railing away from the first door body is detachably and pivotally connected to the second connecting mechanism (3).

13. The new type of sliding door railing according to claim 12, wherein the bottom of the first door body and/or second door body is provided with one or more universal balls for rolling contact with the ground.
